23. Describe the different types of enzyme inhibition mechanisms.

24. Compare competitive and non-competitive inhibition.

25. Write about the various industrial applications of enzymes.

26. Describe the different medical applications of enzymes.

27. Write in detail about the main factors that alter the speed of enzymatic reactions.

SUMMARY

Enzymes are a proteins that work as biological catalysts. They catalyse the biochemical reactions that are occurring in the living organisms. They are highly specific for a given reaction and accelerate the reaction rates to many folds. They help breakdown the food molecules to harvest energy from it and also to synthesise the necessary macromolecules for the development of living organisms. The enzymes have their maximum activity at a particular pH and temperature which are called the optimum pH and temperature for that enzyme. The concentration of the enzyme and the substrate concentration influences the activity of a particular enzyme. Some enzymes do require small organic molecules or metal ions in order to carry out their functions, these molecules are called co enzymes.

Enzymes are classified into six classes based on their functions viz. oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases. The International Union of Biochemistry and Molecular Biology (IUBMB) have assigned a 4 digit number called Enzyme Commission (EC) number.

The activity of an enzymes can be inhibited by small molecules which are known as inhibitors. Based on the type of inhibition they can be classified as competitive, uncompetitive and noncompetitive inhibitors.

The enzymes have many applications in industries as well as in medicine.

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